

Wireless Communications Using MATLAB

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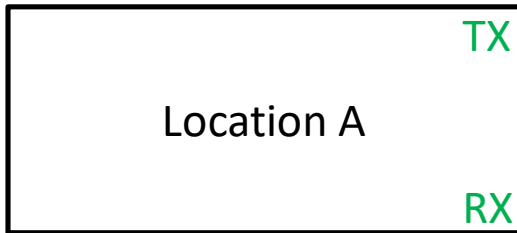
Course Outline

- Simulate a Basic Digital Communications Link
- Implement the essential components of a single-carrier digital communications system, including additive white Gaussian noise.
- Pulse-Shaping Filters
- Incorporate transmit and receive filters into your simulation.
- Multipath Channels, model a multipath channel and evaluate the impact on link quality.
- OFDM, implement orthogonal frequency-division multiplexing (OFDM), a multicarrier modulation scheme

Communication: Exchange of information; Transfers information from one place to another

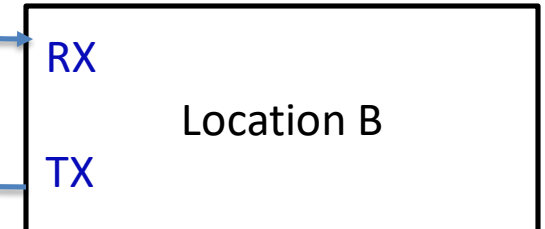
Channel (transmission media)

- Guided (wire)
- Unguided (wireless)



Information/data

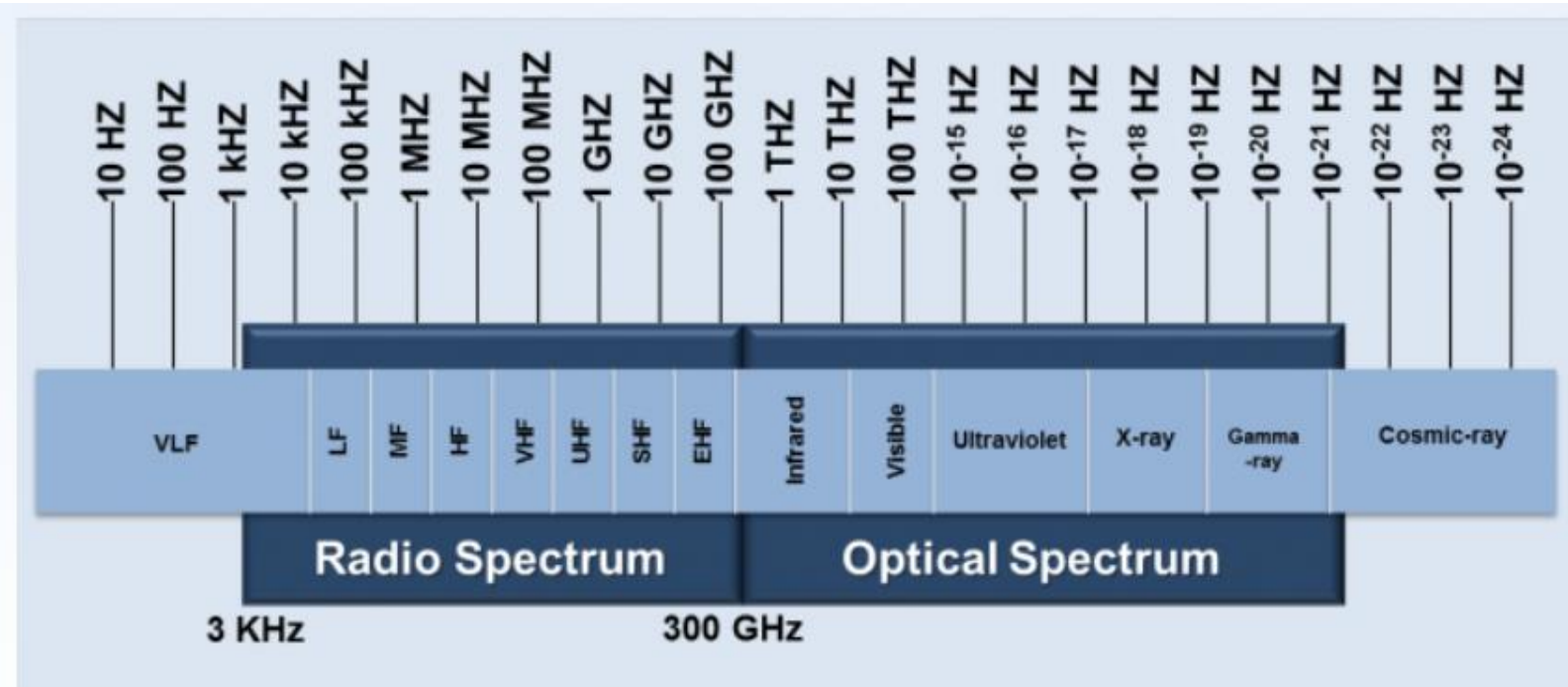
- Audio
- Text
- Video



Communication channel types: different transmission media can be classified based on bandwidths

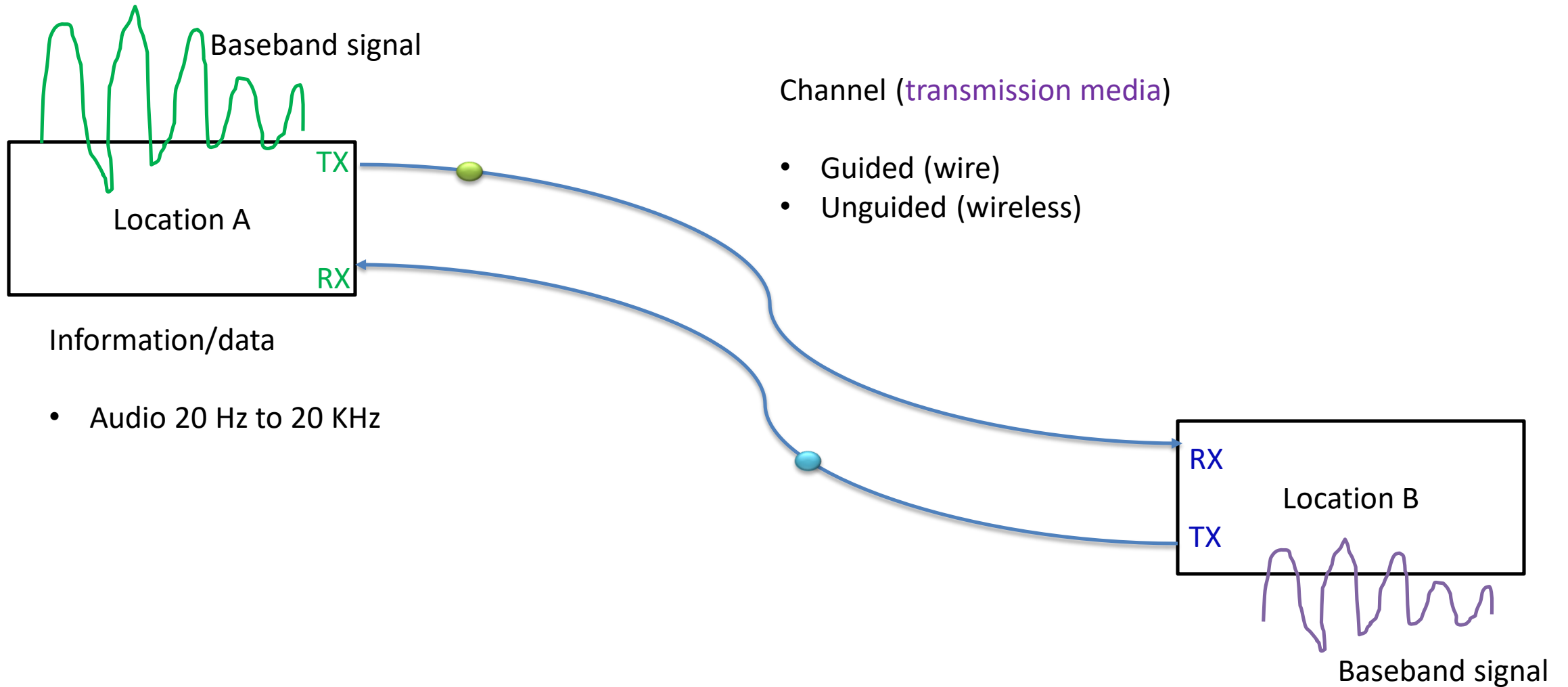
- Guided (wired)----physical media
 - Twisted pair
 - Coaxial cable
 - Optical fiber
- Unguided (wireless)---unbounded media
 - Terrestrial microwave
 - Satellite microwave
 - Broadcast radio
 - Cellular radio
 - Infrared

Electromagnetic Spectrum

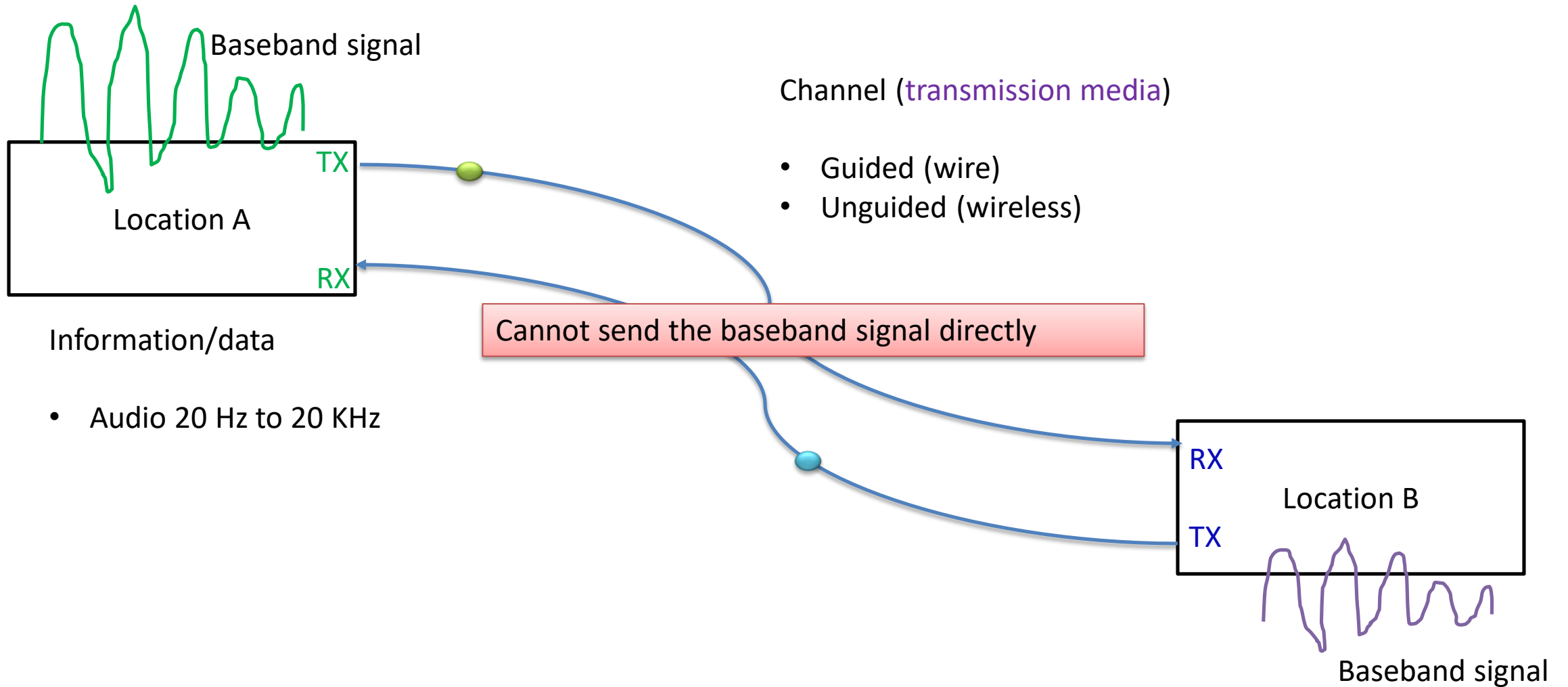


The electromagnetic spectrum is comprised of Radio and Optical Spectrum.

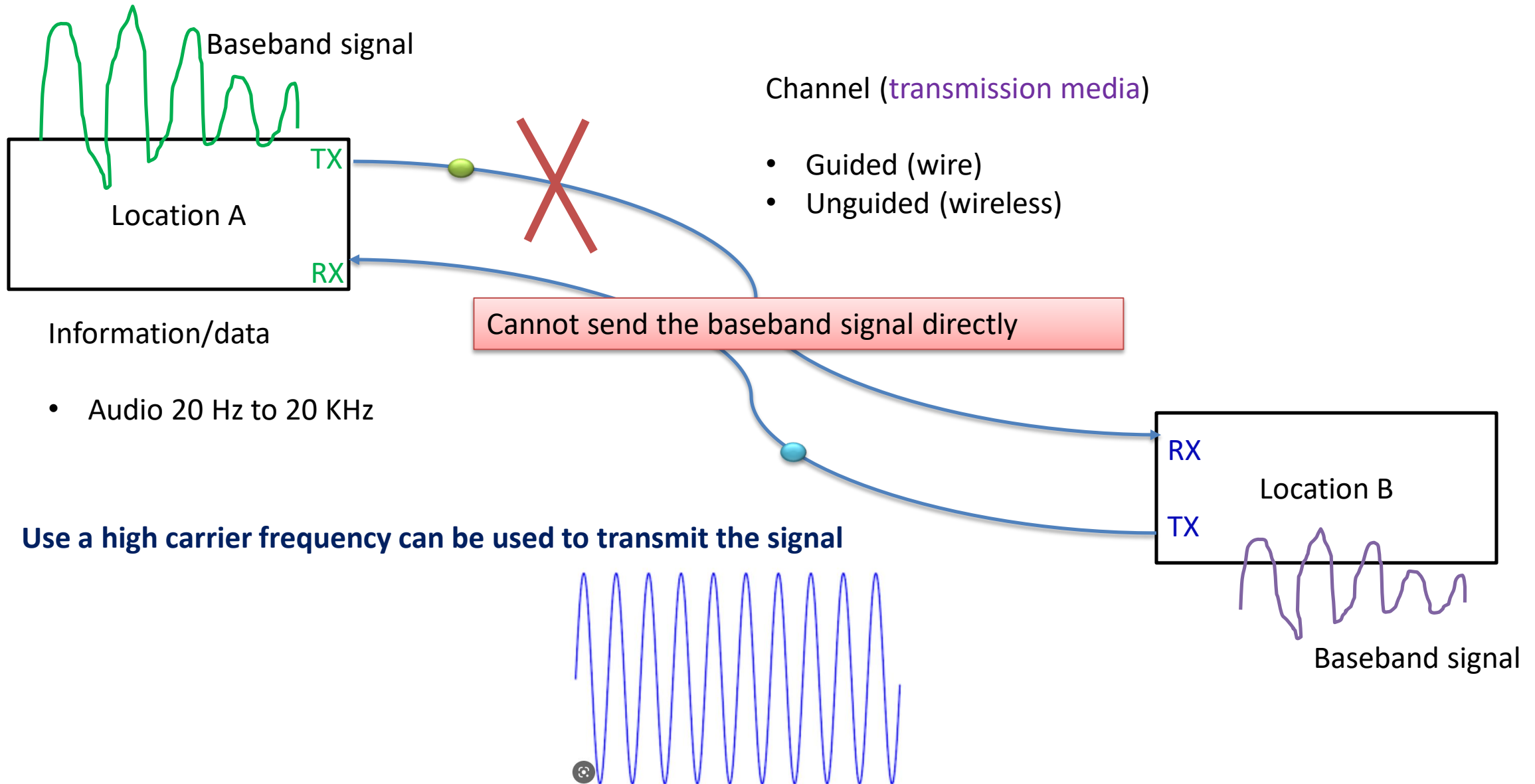
Communication: Voice signal up to 3 KHz



Communication: Voice signal up to 3 KHz

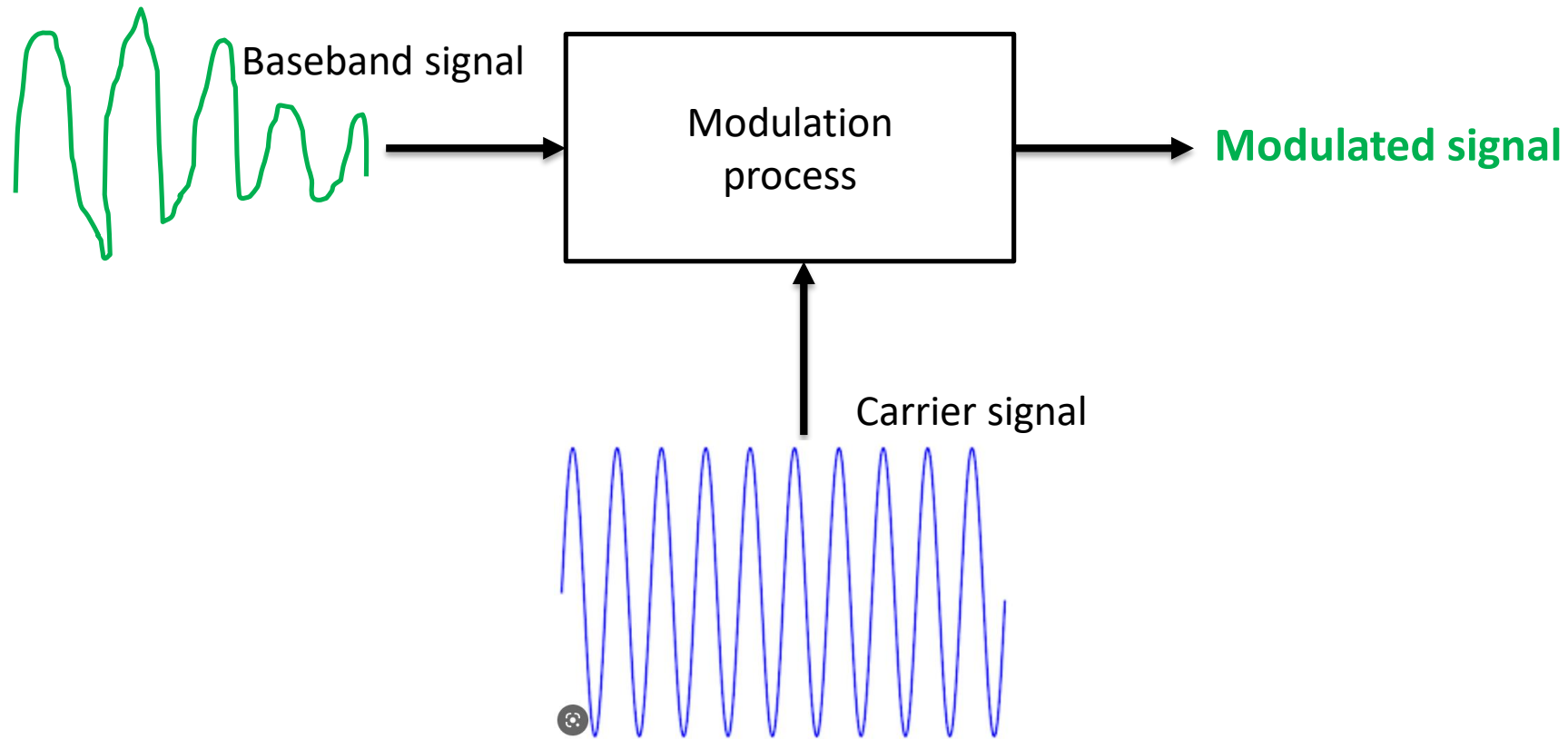


Baseband signal cannot be transmitted through a channel directly due to some challenges?



How to impose the baseband signal into the carrier signal to be transmitted?

- Carrier signal property can be changed based the baseband signal via modulation
- Modulated could change the carrier signal amplitude, frequency, and\or phase based on the baseband signal



Why modulation is used in wireless or wires channels?

- Reduce the size of an antenna

“The length of the antenna is **inversely proportional to the frequency** and directly proportional to the wavelength. The higher the frequency and the shorter the wavelength, the shorter the antenna can be made.” Ref: [nicerf](#)



The advanced technologies in digital electronics reduces the size the performance of an antenna

“Traditionally one antenna works with either one or a few different frequencies. Now we can take advantage of [advanced digital electronics](#) and combine several small antenna elements to work together as one antenna that can be made to [operate digitally](#) with any frequency.” [explains](#) doctoral candidate Jari-Matti Hannula.



Communication system design main factors

- Bandwidth----data rate
 - The amount of data that can be carried through channel per a unit of time
 - Signals are used to transmit data per second in units of Hz
 - In Digital communication system, units is measured based on bits per second (bps)
- Transmission impairment---attenuation & noise
- Number of transmitters & receivers
- System performance & complexity

Problem 1: Find length of antenna for a size of antenna $\lambda/4$

Case A) Frequency $f = 10 \text{ KHz}$

$$C = \lambda * f$$

$$\lambda = c/f = 3 * 10^8 / 10^4$$

$$\lambda = 30000 \text{ m}$$

$$\text{Size of antenna} = \lambda/4 = 7500 \text{ m}$$

Case B) Frequency $f = 300 \text{ MHz}$

$$C = \lambda * f$$

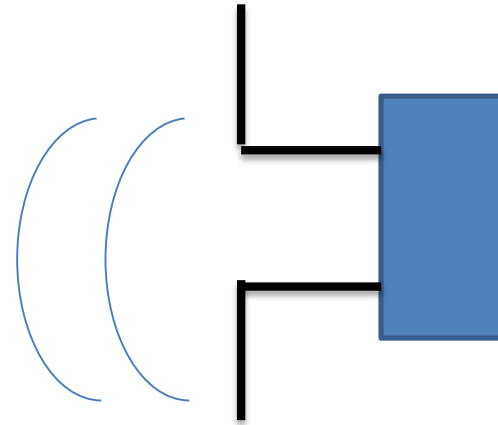
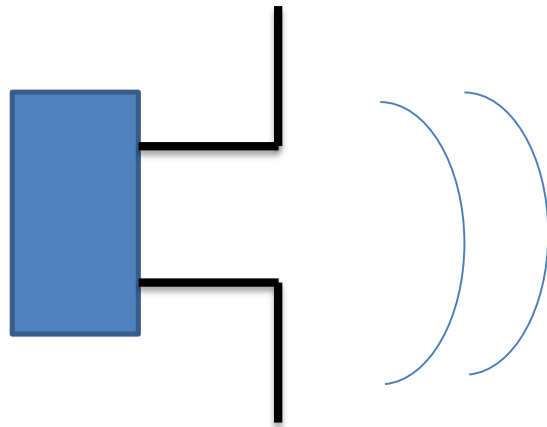
$$\lambda = c/f = 3 * 10^8 / 300 * 10^6$$

$$\lambda = 1 \text{ m}$$

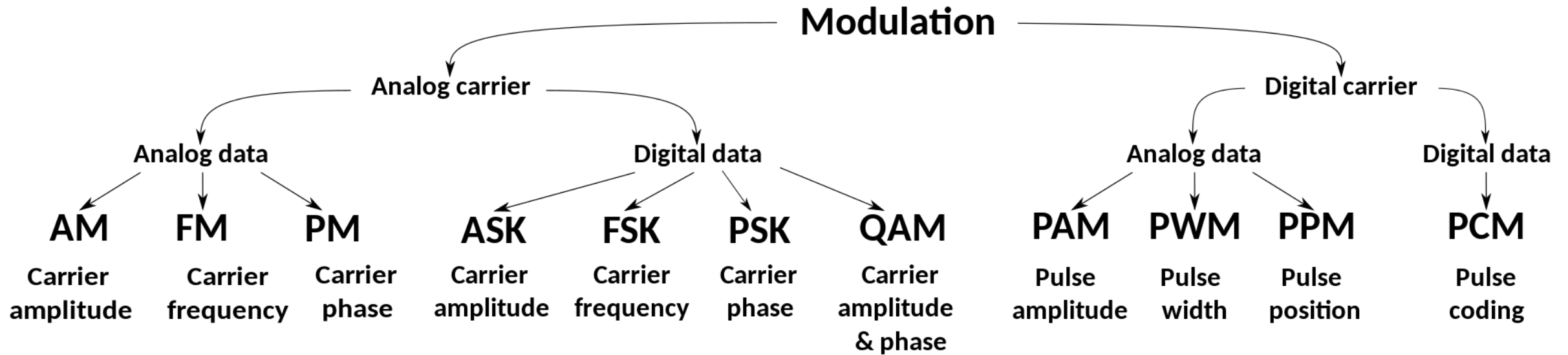
$$\text{Size of antenna} = \lambda/4 = 0.25 \text{ m}$$

Why modulation is used in wireless or wires channels?

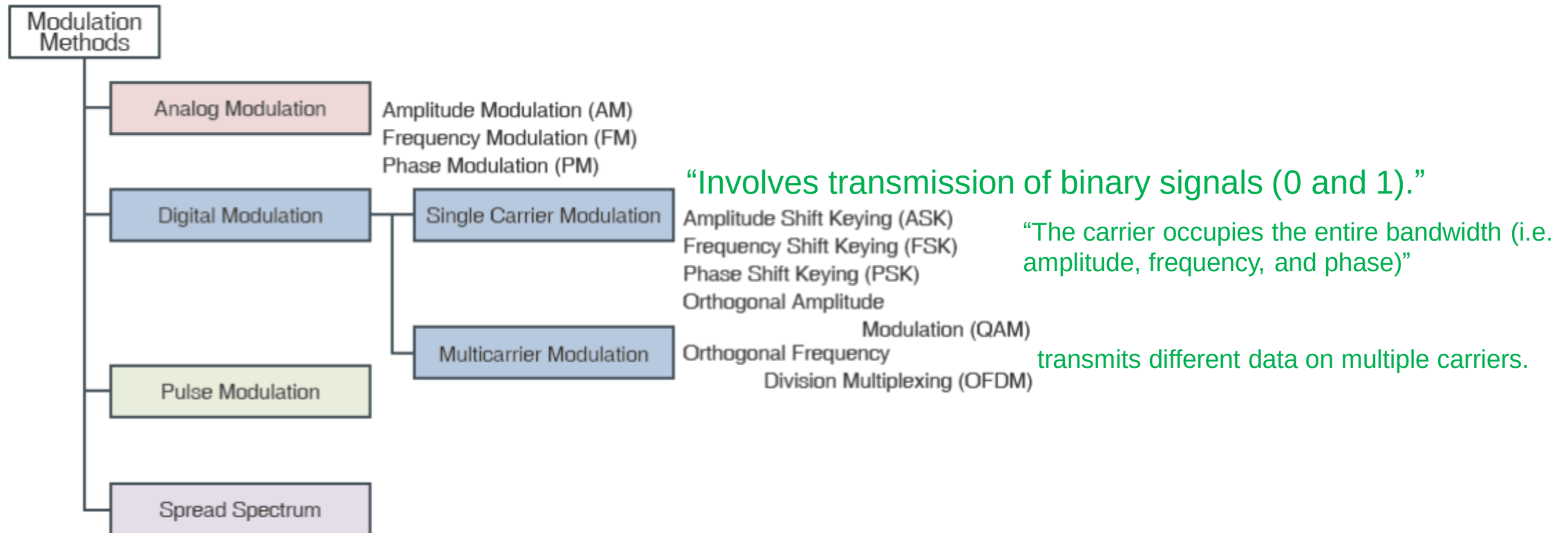
- Reduce the size of an antenna
- Reduce interference between baseband signal
- Transmit multiple signals simultaneously over a single channel for long distance



Modulation types based on data and carrier types



Modulation Method Classification



Signals in Communication system

Signals can be in the form of

- Electromagnetic wave---e.g. light\optical signals through fiber cables
- Electrical signals--e.g. electrical signals through wire cables, needs conductor

A signal is used for carrying data from one location to another

References

- [1] <https://www.tutorialsmate.com/2021/11/difference-between-data-and-information.html>
- [2] <https://physics.stackexchange.com/questions/314695/electromagnetic-wave-vs-electric-current#:~:text=Electric%20current%20is%20the%20movement,can%20propagate%20through%20a%20vacuum.>